

REGULATORY SCORECARD

AI Diabetic Retinopathy Screening Tool

Class C · AI-CDS · MD-7 (Central) · likely MD-22 / test-licence (MD-13) path

LIKELY CLASSIFICATION

Class C · AI-CDS

Moderate complexity

CONFIDENCE

Medium confidence



LIKELY PATHWAY

**MD-7 (Central) · likely MD-22 /
test-licence (MD-13) path**

Clinical investigation likely required.

TIMELINE + COST BAND

12–18 months

Likely compliance spend: ₹17–39 L

READINESS

●●●●●●●●●● 2 / 10

Band: Red (significant work ahead)

RISK LEVEL

High

Risk = potential clinical/regulatory exposure today.

WHAT TRIGGERED THIS CLASSIFICATION

AI-assisted diagnosis

Influences clinical management

Sensitive patient data

Adaptive AI model

TOP GAPS TO CLOSE

No ISO 13485 / IEC 62304 QMS detected

No CDSCO engagement; ACP required for AI/ML SaMD

Limited prospective clinical validation

RECOMMENDED NEXT ACTION

Engage an ISO 13485 + IEC 62304 consultant for a gap assessment and start a QMS build-out plan this quarter.

SECTION 2

Your Likely Regulatory Pathway

Why this likely class applies to your product — and the form sequence reviewers typically expect.

Your product is an AI-driven screening tool that interprets retinal images to flag diabetic retinopathy — a serious, sight-threatening condition where delayed detection can lead to irreversible vision loss. Because the software informs clinical management decisions (refer / monitor / treat) rather than simply logging data, the information it generates carries high significance for patient outcomes. That combination — a serious clinical state paired with software that directly drives care decisions — typically places this kind of AI diagnostic SaMD in the higher-risk band under the Oct 2025 CDSCO SaMD draft, likely Class C. Based on published guidance, the file would likely route through MD-7 via the Central Licensing Authority, with a clinical investigation under MD-22 preceding it.

AUTHORITY

CDSCO Central Licensing Authority

FORMS

MD-12

MD-13

MD-22

MD-7

MD-9

LIKELY STEP SEQUENCE

01 Test licence (MD-12 / MD-13)

Apply via MD-12 for a test licence (granted as MD-13) so a small number of devices can be made for the clinical investigation.

2–3 MONTHS

TEST LICENCE NOTE

A test licence (granted as MD-13, applied via MD-12) typically precedes the clinical investigation. The investigation itself runs under MD-22 / MD-23 permissions before the device can be cleared for commercial manufacture.

02 MD-22 clinical-investigation permission + EC approval

Design the multi-centre study, get EC clearance, register on CTRI, enrol participants.

9–14 MONTHS

03 MD-7 manufacturing licence (CLA → MD-9)

File MD-7 with the Central Licensing Authority alongside QMS evidence, the technical file, predicate or substantial-equivalence narrative, and risk file.

6–9 MONTHS

04 ACP / PCCP submission

File the Algorithm Change Protocol alongside the manufacturing licence — modification scope, retraining triggers, validation thresholds, human oversight.

IN PARALLEL

ACP NOTE

Adaptive AI/ML behaviour means an Algorithm Change Protocol (ACP / PCCP) typically accompanies the MD-7 file per the Oct 2025 CDSCO SaMD draft.

SECTION 3

Readiness Gap Analysis

The gaps a reviewer is likely to flag, ordered by priority. Effort and cost bands are typical Indian medtech ranges — your numbers may be lower with in-house resources or higher with a Big-Four-style consultant.

PRI	GAP	WHY IT MATTERS	SUGGESTED NEXT STEP	EFFORT · COST
P1	No ISO 13485 / IEC 62304 QMS detected	Reviewers typically expect ISO 13485-aligned QMS evidence — design controls, CAPA, complaint handling, internal audit records, and IEC 62304 software lifecycle artefacts — before they engage on a Class C AI-CDS submission. For an AI/ML diagnostic like a DR screening tool, the QMS also anchors how training data, model versioning, and post-market monitoring are governed. Without a documented quality system in place, the submission package is likely to be held at intake and any clinical investigation approval may be delayed until QMS maturity can be demonstrated.	Engage an ISO 13485 + IEC 62304 consultant for a gap assessment and start a QMS build-out plan this quarter.	6–9 months · ₹3–5 L
P1	No CDSCO engagement; ACP required for AI/ML SaMD	Based on the Oct 2025 CDSCO SaMD draft guidance, adaptive AI/ML devices are typically expected to file an Algorithm Change Protocol (ACP) — or a Predetermined Change Control Plan — that describes the scope of permitted model modifications, retraining triggers, validation thresholds, rollback criteria, and the human oversight model. For a DR screening tool where sensitivity and specificity can drift with new training data, this is what lets you push model updates post-approval without re-filing each time. Without an ACP framework prepared early, every meaningful model change may require a fresh submission, which can stall your product roadmap.	Open a CDSCO pre-submission consultation track and draft an Algorithm Change Protocol (ACP) for the retraining lifecycle.	3–5 months · ₹2–4 L

P1

Limited prospective
clinical validation

For a Class C AI diagnostic with a novel indication, reviewers will likely require prospective Indian-population clinical evidence — typically a multi-centre study with Ethics Committee approval and CTRI registration — to confirm that performance generalises across local retinal imaging conditions, cameras, and patient demographics. Retrospective dataset accuracy alone is rarely accepted for autonomous or assistive DR screening. This is usually the longest item on the critical path, so starting site selection, EC submissions, and CTRI registration early is the single biggest lever on your overall timeline to market.

Design an EC-approved prospective validation study on Indian retinal datasets with ophthalmologist-adjudicated ground truth; register on CTRI.

9–14 months · ₹8–18 L

SECTION 4

Timeline + Cost Estimator

Phased roadmap with Indian-context cost bands. Total range and anchor reflect the readiness card's headline estimate.

TOTAL **12–18 months** Anchored to typical Class C AI-SaMD timelines: QMS build-out (~4–6 months), clinical validation with EC approval (~6–12 months), and central CDSCO review (~3–6 months).

Phase 1 — Foundational compliance

3–6 MONTHS ₹3–6 L

Engage an ISO 13485 consultant for a gap assessment, document the Intended Use Statement, and lock the classification rationale.

Phase 2 — Technical documentation

4–6 MONTHS ₹3–7 L

Build the IEC 62304 software lifecycle file, ISO 14971 risk file, IEC 81001-5-1 cybersecurity controls, and IEC 62366-1 usability evidence.

Phase 3 — Clinical validation

9–14 MONTHS ₹8–18 L

Design and run a multi-centre clinical investigation with EC approval and CTRI registration. Typically the longest item on the critical path.

Phase 4 — Submission preparation

2–3 MONTHS ₹2–5 L

Assemble the Device Master File, labelling, IFU, predicate or SE narrative, and pre-submission review.

Phase 5 — CDSCO review cycle

4–8 MONTHS ₹1–3 L

Submission filing, queries-and-responses cycle, possible site inspection, licence grant.

LIKELY BOTTLENECKS

- Clinical investigation timeline (EC + CTRI + enrolment + follow-up) dominates the schedule.
- Top high-severity gap: No ISO 13485 / IEC 62304 QMS detected.
- ACP filing introduces additional review back-and-forth alongside the main licence.

SECTION 5

Reviewer Insights

What CDSCO reviewers will likely look for in a submission of this shape. Generic regulatory checklists rarely capture reviewer psychology — this is yours.

Strength of clinical validation evidence

Reviewers likely expect prospective Indian-population validation on fundus images spanning diverse cameras, skin tones, and DR severity grades (ICDR 0-4), with EC approval and CTRI registration in place before enrolment. Sensitivity and specificity against a masked ophthalmologist-adjudicated reference standard, pre-specified non-inferiority margins, and subgroup analyses by age and diabetes duration typically receive line-by-line scrutiny for a Class C AI-CDS.

Intended Use consistency across documents

Reviewers typically cross-check whether the tool is positioned as a screening aid for referable DR or as a standalone diagnostic. Drift between marketing copy ("AI detects diabetic retinopathy"), the IFU (decision support for trained clinicians), and labelling on intended user, care setting, and image-quality prerequisites may trigger early queries. Patient population, contraindications, and the human-in-the-loop caveat likely need to read identically across all artefacts.

AI/ML update controls (ACP / PCCP)

Per the Oct 2025 CDSCO SaMD draft, reviewers likely expect an Algorithm Change Protocol covering retraining triggers (drift in referable-DR sensitivity, new camera onboarding, population shift), locked validation thresholds, shadow-mode evaluation, and rollback criteria. The boundary between pre-approved changes and those needing fresh notification typically needs explicit definition, with human oversight and ophthalmologist sign-off documented before any commercial model update.

Cybersecurity & data handling

Since retinal images are identifiable health data, reviewers typically ask for a threat model aligned to IEC 81001-5-1, AES-256 encryption at rest and TLS 1.2+ in transit, role-based access for screening operators and reviewing ophthalmologists, audit logging, and a breach-notification SOP. For cloud-hosted inference, CERT-In empanelled audit reports, data-residency statements, and DPDP Act 2023 alignment for consent and retention likely receive close attention.

Explainability of AI recommendations

For an AI-CDS influencing referral decisions, reviewers typically expect the output to show a referable/non-referable call with a calibrated confidence score, heatmaps or lesion-level overlays highlighting microaneurysms, haemorrhages, or neovascularisation, and an explicit ungradable-image flag. Uncertainty handling for borderline cases and clear messaging that the ophthalmologist remains the responsible decision-maker likely need to be visible in the UI and reflected in the IFU.

SECTION 6

Smart Examples

Annotated good-vs-bad wording. These are snippets, not full forms — copy them as models for your own Intended Use, claim, and risk-justification language.

INTENDED USE **Scoped vs. autonomous Intended Use**

GOOD

The device assists ophthalmologists and trained graders by flagging colour fundus photographs with features suggestive of referable diabetic retinopathy (moderate NPDR or worse) and DME, presented as an advisory output for clinician review in screening clinics. It does not perform autonomous diagnosis; the ophthalmologist or designated grader remains the responsible decision-maker for referral.

BAD

The device automatically diagnoses diabetic retinopathy from fundus photographs and issues the final referral decision without clinician review.

WHY THIS IS SAFER

The good wording scopes the tool as advisory, names the clinician as decision-maker, and bounds the indication to referable DR and DME — aligning with the Oct 2025 CDSCO SaMD draft. The bad wording claims autonomous diagnosis, typically escalating class and oversight scrutiny.

CLAIM WORDING **Performance claim with study basis**

GOOD

In a retrospective evaluation on a held-out test set of [N] fundus images from [centre], graded against a reference standard of [3 retina specialists, majority vote], the device demonstrated [sensitivity]% sensitivity and [specificity]% specificity for referable DR at operating threshold [t]. Performance on prospective Indian screening-population data across varied camera models is pending.

BAD

The device matches ophthalmologist-level accuracy in detecting diabetic retinopathy from retinal images.

WHY THIS IS SAFER

The good wording names study type, sample size, reference standard, operating point, and pending prospective work. The bad wording uses an unverifiable comparator and an unbounded claim. Reviewers typically ask for the underlying study report before accepting any performance number.

RISK JUSTIFICATION **False-negative risk justification**

GOOD

Hazard: missed referable diabetic retinopathy or DME.
Hazardous situation: screener defers referral based on a negative device output and the patient is not sent for ophthalmology review. Harm: delayed treatment, progression to vision-threatening DR.
Mitigation: output framed as advisory; protocol requires periodic re-screening and clinician oversight of negatives; operating point tuned to favour sensitivity (typically published sensitivity $\geq$ X% for referable DR); ungradable images flagged for repeat capture; residual risk tracked under PMS.

BAD

The model is highly accurate on retinal images, so missed DR cases are unlikely in practice.

WHY THIS IS SAFER

The good wording follows ISO 14971's hazard-situation-harm chain, names mitigations including ungradable-image handling and sensitivity-favoured thresholding, and acknowledges residual risk under PMS. The bad wording handwaves false negatives and leaves reviewers without an auditable trail.